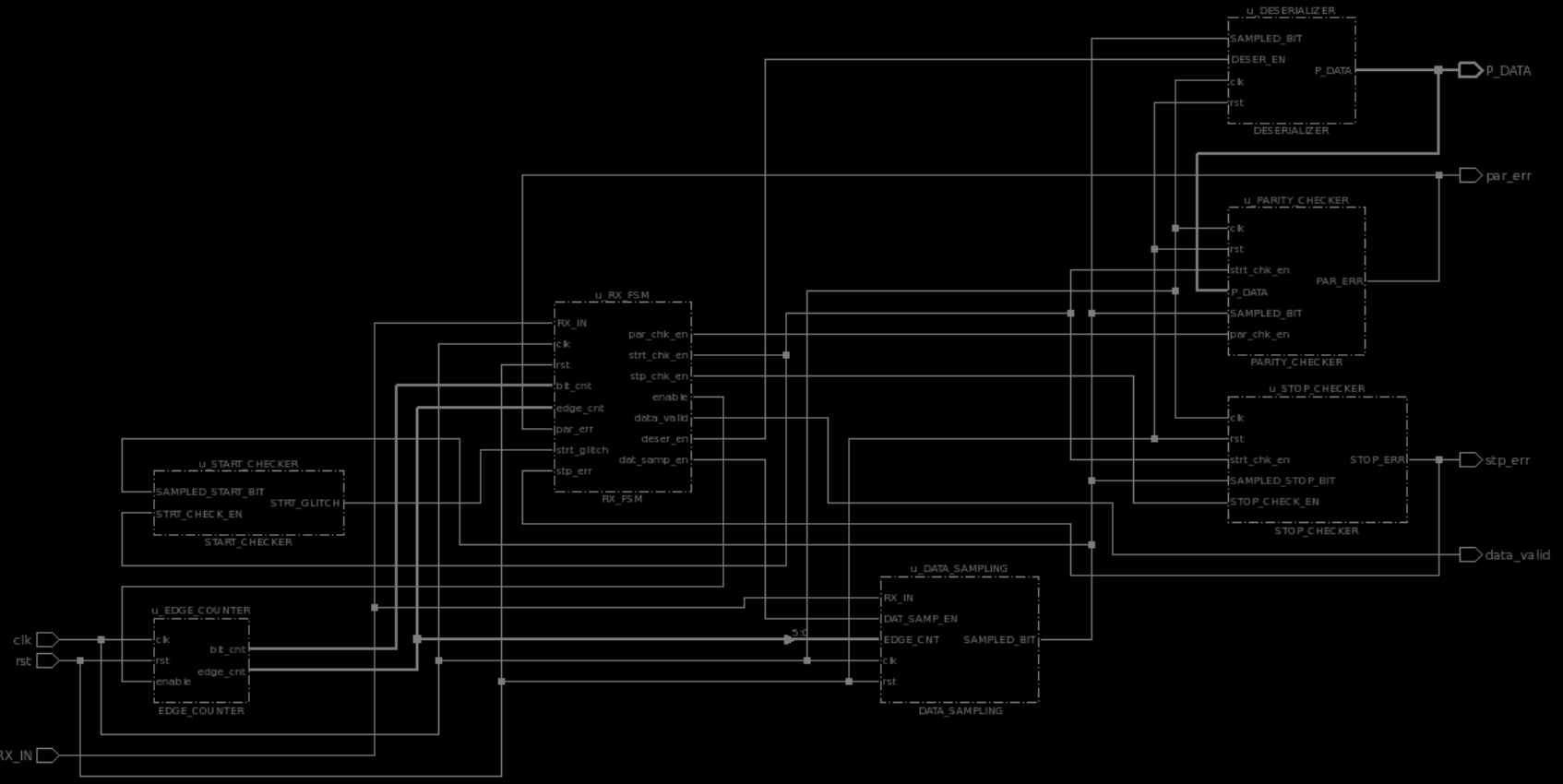
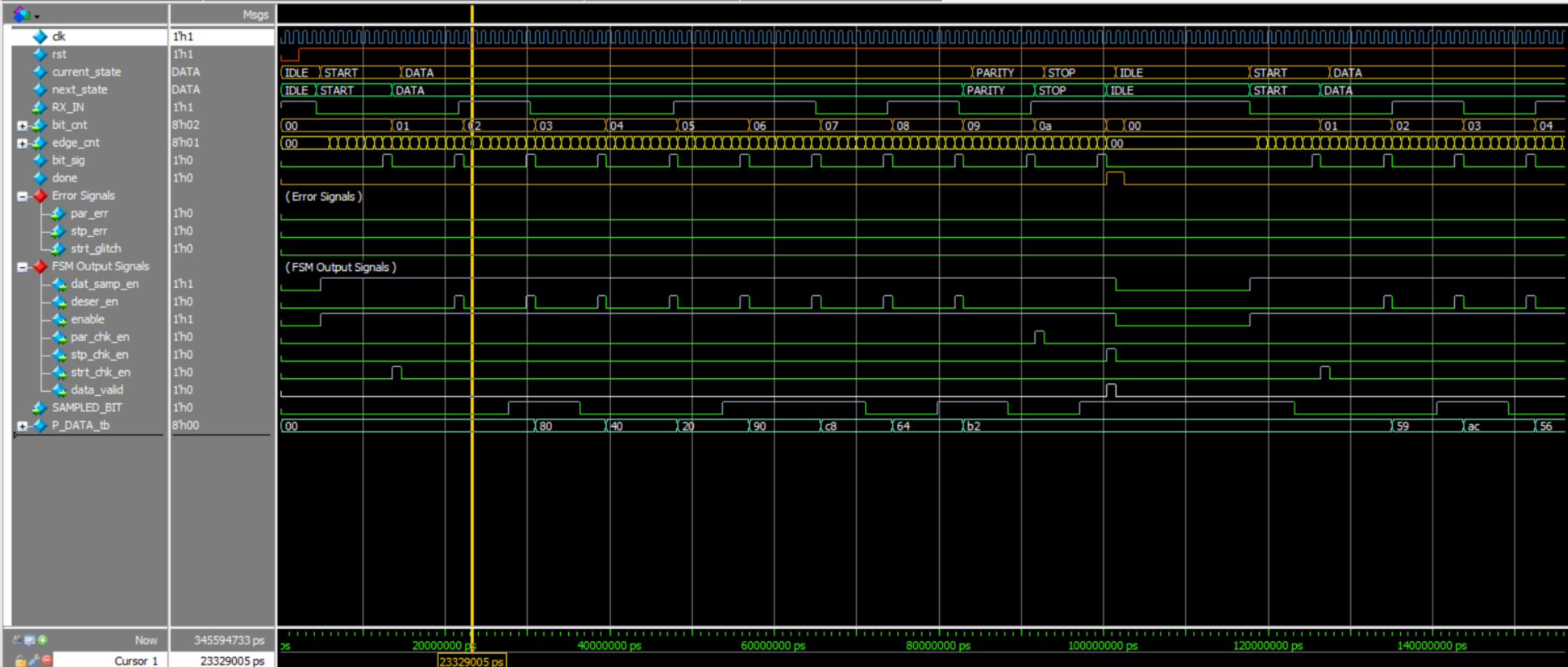

u_DESERIALIZER
SAMPLED_BIT
DESER_EN
clk
rst
P_DATA
DESERIALIZER
P_DATA
par_err
u_PARITY_CHECKER
clk
rst
strt_chk_en
P_DATA
SAMPLED_BIT
par_chk_en
PAR_ERR
PARITY_CHECKER
u_RX_FSM
RX_IN
clk
rst
bit_cnt
edge_cnt
par_err
strt_glitch
stp_err
par_chk_en
strt_chk_en
stp_chk_en
enable
data_valid
deser_en
dat_samp_en
RX_FSM
u_STOP_CHECKER
clk
rst
strt_chk_en
SAMPLED_STOP_BIT
STOP_CHECK_EN
STOP_ERR
STOP_CHECKER
stp_err
data_valid
u_START_CHECKER
SAMPLED_START_BIT
STRT_CHECK_EN
STRT_GLITCH
START_CHECKER
u_DATA_SAMPLING
RX_IN
DAT_SAMP_EN
EDGE_CNT
clk
rst
SAMPLED_BIT
DATA_SAMPLING
u_EDGE_COUNTER
clk
rst
enable
bit_cnt
edge_cnt
EDGE_COUNTER
clk
rst
RX_IN


Msgs
clk
1'h1
rst
1'h1
current_state
DATA
next_state
DATA
RX_IN
1'h1
bit_cnt
8'h02
edge_cnt
8'h01
bit_sig
1'h0
done
1'h0
Error Signals
par_err
1'h0
stp_err
1'h0
strt_glitch
1'h0
FSM Output Signals
dat_samp_en
1'h1
deser_en
1'h0
enable
1'h1
par_chk_en
1'h0
stp_chk_en
1'h0
strt_chk_en
1'h0
data_valid
1'h0
SAMPLED_BIT
1'h0
P_DATA_tb
8'h00
IDLE
START
DATA
PARITY
STOP
IDLE
START
DATA
(Error Signals)
(FSM Output Signals)
00
01
02
03
04
05
06
07
08
09
0a
00
01
02
03
04
00
00
80
40
20
90
c8
64
b2
59
ac
56
Now
345594733 ps
Cursor 1
23329005 ps
ps
20000000 ps
40000000 ps
60000000 ps
80000000 ps
100000000 ps
120000000 ps
140000000 ps
23329005 ps

```
sim:/UART_TOPR_TB/P_DATA_tb
VSIM(paused)> run -all
# [PASS] Frame 1: 8'hB2 Normal Reception | P_DATA: b2 | par_err: 0 | stp_err: 0 time 100369000
# [PASS] Frame 2: 8'h5A Normal Reception | P_DATA: 5a | par_err: 0 | stp_err: 0 time 213216000
# [PASS] Frame 3: Parity Error Injection | P_DATA: ff | par_err: 1 | stp_err: 0 time 328234000
# ================================================
#    ALL TESTS COMPLETED.
# ================================================
# ** Note: $finish    : UART_TOPR_TB.v(159)
#    Time: 345594733 ps  Iteration: 0  Instance: /UART_TOPR_TB
```